

Calculation status: NLO hard functions

10 September 2026

Process	Order	Channel	Total time per channel	Size [kB]	Verification	Source
$pp \rightarrow hX$ UU	NLO	$qq' \rightarrow q + X$	2.54 min	22.9	✓	INCNLO
		$q\bar{q} \rightarrow q' + X$	2.43 min	25.6	✓	INCNLO
		$qg \rightarrow q + X$		143.3	✓	INCNLO
		$qg \rightarrow g + X$		71.9	✓	INCNLO
$pp \rightarrow hX$ LL	NLO	$qq' \rightarrow q + X$	2.80 min	16.9	✓	Navis[†]
		$q\bar{q} \rightarrow q' + X$	2.65 min	25.7	✓	Navis
		$qg \rightarrow q + X$		76.3	✓	Navis
		$qg \rightarrow g + X$		47.8	✓	Navis
$pp \rightarrow hX$ TT	NLO	$qq' \rightarrow q + X$	26.0 s	1.7	N/A	N/A
$pp \rightarrow hX$ LL transfer	NLO	$qq' \rightarrow q + X$	2.89 min	22.9	N/A	N/A
$pp \rightarrow hX$ TT transfer	NLO	$qq' \rightarrow q + X$	13.57 min	20.5	N/A	N/A
Drell–Yan UU	NLO	$q\bar{q}$	21.5 s	4.4	✓	1307.6925
		$\bar{q}q$	19.3 s	4.4	✓	1307.6925
		qg	8.5 s	3.0	✓	1307.6925
		gq	5.7 s	3.0	✓	1307.6925
		$\bar{q}g$	6.1 s	3.0	✓	1307.6925
		$g\bar{q}$	5.9 s	3.0	✓	1307.6925
SIDIS UU	NLO	$q \rightarrow q$	26.7 s	10.4	✓	2401.16281
		$\bar{q} \rightarrow \bar{q}$	25.0 s	10.4	✓	2401.16281
		$q \rightarrow g$	16.8 s	5.6	✓	2401.16281
		$\bar{q} \rightarrow g$	18.5 s	5.6	✓	2401.16281
		$g \rightarrow q$	7.7 s	4.9	✓	2401.16281
		$g \rightarrow \bar{q}$	9.5 s	4.9	✓	2401.16281
SIDIS LL	NLO	$q \rightarrow q$	21.5 s	9.8	✓	2404.08597
		$\bar{q} \rightarrow \bar{q}$	23.2 s	9.8	✓	2404.08597
		$q \rightarrow g$	18.6 s	5.2	✓	2404.08597
		$\bar{q} \rightarrow g$	17.2 s	5.2	✓	2404.08597
		$g \rightarrow q$	6.5 s	4.2	✓	2404.08597
		$g \rightarrow \bar{q}$	4.8 s	4.2	✓	2404.08597

Notes. Times are historical complete production-driver runtimes, excluding startup and independent reference comparisons; an empty cell means no complete runtime is recorded. Sizes are final Mathematica files, with 1 kB = 1000 bytes. ✓ means the full finite coefficient comparison passed; N/A means no retained full external comparison.

For $pp \rightarrow hX$, LL/TT polarize both incoming partons; *transfer* polarizes one incoming and the observed outgoing parton. SIDIS lists incoming $\rightarrow$ observed parton; $q' \neq q$. The incoming-TT qq' result is exactly zero and passes internal checks. [†]Also checked against the supplied Vogelsang Mathematica expressions.

Calculation status: SIDIS NNLO hard functions

10 September 2026

Process	Order	Channel	Total time per channel	Size [kB]	Verification	Source
SIDIS UU	NNLO	$q \rightarrow q$		850.5	✓	2401.16281
		$\bar{q} \rightarrow \bar{q}$		850.5	✓	2401.16281
		$q \rightarrow g$		208.7	✓	2401.16281
		$\bar{q} \rightarrow g$		208.7	✓	2401.16281
		$g \rightarrow q$		227.0	✓	2401.16281
		$g \rightarrow \bar{q}$		227.0	✓	2401.16281
		$q \rightarrow \bar{q}$		159.8	✓	2401.16281
		$\bar{q} \rightarrow q$		159.8	✓	2401.16281
		$q \rightarrow q'$		106.5	✓	2401.16281
		$q \rightarrow \bar{q}'$		106.6	✓	2401.16281
		$\bar{q} \rightarrow q'$		106.6	✓	2401.16281
		$\bar{q} \rightarrow \bar{q}'$		106.5	✓	2401.16281
		$g \rightarrow g$		95.9	✓	2401.16281
SIDIS LL	NNLO	$q \rightarrow q$		387.3	✓	2404.08597
		$\bar{q} \rightarrow \bar{q}$		387.3	✓	2404.08597
		$q \rightarrow g$		135.9	✓	2404.08597
		$\bar{q} \rightarrow g$		136.0	✓	2404.08597
		$g \rightarrow q$		119.2	✓	2404.08597
		$g \rightarrow \bar{q}$		119.2	✓	2404.08597
		$q \rightarrow \bar{q}$		93.6	✓	2404.08597
		$\bar{q} \rightarrow q$		93.6	✓	2404.08597
		$q \rightarrow q'$		60.4	✓	2404.08597
		$q \rightarrow \bar{q}'$		60.4	✓	2404.08597
		$\bar{q} \rightarrow q'$		60.4	✓	2404.08597
		$\bar{q} \rightarrow \bar{q}'$		60.4	✓	2404.08597
		$g \rightarrow g$		48.3	✓	2404.08597

Notes. All real, virtual and UV/PDF/FF counterterms are included, with the finite helicity-scheme conversion for LL. The 26 final files total 5.18 MB and contain no unknown constants or unevaluated integrals.

Every delta, plus and regular coefficient passes independent numerical comparisons at five parameter points, with approximately 40-digit agreement. The master calculation and endpoint analysis are shared across channels; complete individual-channel runtimes are not recorded.